



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid-Term Exam Trimester: Spring 2024

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers

Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (6 Marks)

Transfer of bus control from processor to device takes **300 ns**. Transfer of bus control from device to processor takes **200 ns**.

a.	If one of the input/output devices employs DMA in cycle stealing mode and takes 12564000 ns to transfer 512 bytes of data. If data is transferred 4 bytes at a time, what is the data transfer rate of the device in KB/s ?	[3]
b.	Suppose, you are transferring 8192 bytes of data. Which mode(burst mode/cycle stealing mode) will be faster to transfer this data? Assume that in cycle stealing mode, data is transferred 4 bytes at a time . ? (<u>Use the data transfer rate found from (a)</u>)	[3]

Question 2: Answer all the questions. (6 Marks)

a.	Suppose, execution of a signed subtraction instruction (8000H - 0001H) occurred, What would be the value of carry flag (CF), sign flag (SF), parity flag (PF), overflow flag (OF)?	[4]
b.	Proof mathematically, the maximum size of a segment is 64KB in an 8086 microprocessor.	[2]

Question 3: Answer all the questions. (6 Marks)

a.	SS = 2526H BX = 0020H BP = 1105H CS = A231H To access the physical address in the Stack Segment , what should be the value of I) the segment register if the offset register holds the value of 3C5AH. II) the offset register if the segment register holds the value of 213AH. Justify whether your answer is valid or not. III) Find the first and last physical address of the Code Segment .	[3]
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b.	RAM[A] has a data bus of 16 bits and RAM[B] has an address bus of 17 bits. Both RAMs have a total memory capacity of 128 KB. What is the address bus width of RAM[A] and data bus width of RAM[B]?	[3]
<u>Question 4: Answer all the questions. (6 Marks)</u>		
a.	Draw the diagram of a microprocessor with a 20-bit address bus and 8-bit data bus interfaced to 8 KB RAM system using the full decoding method . Each RAM chip has a 11-bit address bus and 8-bit data bus. Draw the circuit to address memory range 58000H – 59FFFH . Provide the address mapping as well.	[6]

<u>Question 5: Answer all the questions. (6 Marks)</u>		
a.	In the interrupt vector table, the upper byte of CS for the ultrasonic motion detector is in the physical address 0036BH , and the upper byte of IP for the fire alarm sensor is in the physical address 001E5H . I) Determine each sensor's interrupt number in hexadecimal. II) If both sensors send interrupt signals to the microprocessor at the same time, which one will be executed first and why?	[4]
b.	If there are a total 128 interrupts in an 8086 microprocessor then what will be the size of the interrupt vector table ?	[2]